DANIEL RUIZ FIGUEROA

COMPUTER ENGINEER

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PROFILE

Computer Engineer based in Madrid, working as a Calypso developer in capital markets and heading for the space sector. I built SKYFALL, an open-source tool that maps where re-entering space debris is likely to fall, and a spacecraft descent and landing simulator as my final degree project.

EXPERIENCE

Calypso Developer

Jan 2025 – Present · Madrid

Qaracter · job title: Financial Analyst

- Develop and maintain trade confirmations for FX products on the Calypso platform.
- Generate the confirmation documents with the specific data each product requires.
- Manage delivery across confirmation channels: email, fax, TradeServ and SWIFT.

`Calypso` `Java` `Oracle SQL`

PROJECTS

SKYFALL

2026

Re-entry risk maps for space debris · open source (MIT)

- Turns official re-entry warnings (TIP messages from Space-Track) into 90, 95 and 99 % probability zones on a 3D globe, and compares them with the airspace closure strips from the EASA bulletins.
- Samples re-entry times from the measured error of a year of past warnings (about 4,450 warnings, 680 objects), propagates each orbit with SGP4 and scatters debris with Monte Carlo runs on a 0.5° grid.
- Validated on 100 real warnings the model had never seen: the 95 % zone contained 95 confirmed impact points, against 87 for the 120 km strip, for about the same closed area.

`Python` `FastAPI` `pytest` `React` `TypeScript` `CesiumJS` `developed with Claude Code`

Spacecraft Descent and Landing Simulator

2024

Final degree project · solo · graded 8.9 / 10

- Simulates descent and landing on the Moon, Mars and a configurable planet: gravity, wind, thrust, fuel burn that lightens the vehicle, hull heating, heat shield and parachute.
- Mission parameters are set before each run; vehicle models are streamed at runtime from Microsoft Azure Blob Storage with private access keys, with a local fallback.

`C#` `Unity` `Azure Blob Storage` `REST`

Love Engine

University

Custom 3D engine in C++ · team project

- Engine built on Ogre3D (rendering), Bullet (physics), FMOD (audio) and Lua scripting, with a complete 3D title developed on top of it.

`C++` `Lua` `Ogre3D` `Bullet` `FMOD`

SKILLS

LANGUAGES

Python, C#, C++, C, Java, TypeScript, JavaScript, SQL

SPACE & SIMULATION

SGP4 orbit propagation, Monte Carlo methods, rigid-body physics, Unity, CesiumJS

BACKEND & CLOUD

FastAPI, REST APIs, Microsoft Azure, Oracle SQL, SQLite

FINANCIAL SYSTEMS

Calypso, FX trade confirmations, SWIFT

TOOLS & PRACTICE

Git, pytest, unit testing, Visual Studio, VS Code, Jupyter, Scrum

EDUCATION

BSc in Video Game Development

Universidad Complutense de Madrid

2019 – 2024

Programming in C++ and C#, physics simulation, artificial intelligence and machine learning, networking, and team projects run with Scrum and QA.

LANGUAGES

SPANISH

Native

ENGLISH

B2 · Cambridge First Certificate

OTHER

DRIVING LICENCE

B